AI-IoT/Data Science Take Home Assignment Question

1.

Apart from the usual data analysis and data modelling, data collection and pipelining remains a crucial part of the data lifecycle. Without proper ingestion of healthy data, any downstream tasks would be very ineffective. By extension, this would mean that the relevant software engineering skills for building proper data collection pipelines would be important as well.

In this task, you will be using both docker and docker-compose to build a simple REST API application. Docker is a popular containerization software that can be used to bundle applications such as python applications so that they can be easily deployed in different environments. Docker Compose is a tool to combine one or more docker containers together in an easily deployable stack.

The goal of this task is to create a simple REST API that has an endpoint (/generateReport) that accepts some log data in the form of a csv file from a machine, does some processing on it and returns a report summary of the logs.

Specifications and Deliverables:

- The project should be bundled into a docker-compose file with the relevant Dockerfiles. You are to produce the entire project repository with the relevant source code in a zip file.

- The project should be deployed with the command `docker compose up` in the root folder of the project. It should then deploy a relevant rest api on the localhost.

- For each relevant numerical feature, the API should return its:
    - Min value
    - Max value
    - Mean
    - Standard Deviation
    - Kurtosis

- You may also explain your methodology in the README.md of the project.

- The API should be hosted on port 10000

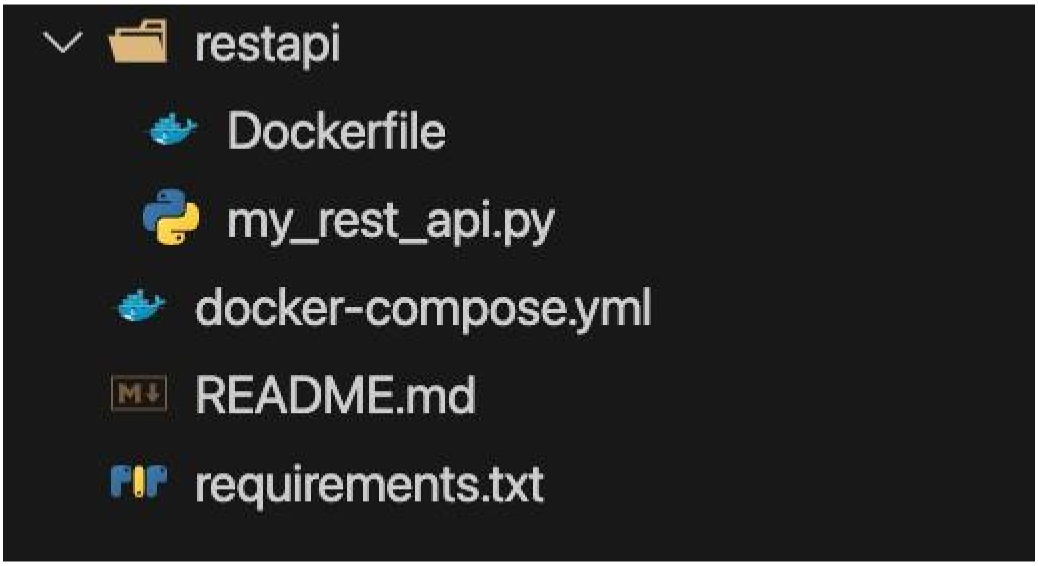
- A sample CSV logs can be accessed here: https://drive.google.com/file/d/11jd6IGZMIkwEW8vlS4jz9sXfctDL5tTK/view?usp=sharing

Hints:

- If you are using Mac or Windows, you can install docker desktop here: https://www.docker.com/products/docker-desktop/

- If you are using Linux, you can install docker here: https://docs.docker.com/engine/install/

- The project structure should look something like this:



- You may use the following code snippet to test if your api works:

```python
import requests


url = 'http://localhost:10000/generateReport'

file_path = 'M6.csv'

# Open the file in binary mode
with open(file_path, 'rb') as file:
    # Define the file dictionary; the key ('file' in this case) is the
name of the form field
    files = {'file': (file_path, file)}
    response = requests.post(url, files=files)
    if response.status_code == 200:
        print("File uploaded successfully.")
        print(response.text)
    else:
```

```
                print(f"Failed to upload file. Status code:
{response.status_code}")
                print(response.text)
```

Question 2.

As machine learning practitioners, datasets are commonly given with ground truth so that we may train our models to fit the distribution of our dataset and "learn" the data. However, in real world scenarios, ground truth labels are rare and may even be incorrectly labelled due to a variety of reasons.

As such we often first need to make sense of the data, this could be by using a combination of methods such as Exploratory Data Analysis(EDA), feature extraction, feature engineering, data cleaning, data visualisation and unsupervised learning.

The ESC-50 Dataset is a dataset of 2000 5 second audio clips of different types of sounds that we can use to explore audio data. In this exercise, your task is to use appropriate feature extraction, visualisation techniques and unsupervised learning techniques to identify patterns in the data. The goal would be to identify patterns in the data that could be used to predict the class of the audio **without relying on ground truth labels**.

Specifications and Deliverables:

- The dataset and details about how to download the dataset can be found here: https://github.com/karolpiczak/ESC-50
- There are 50 classes in the dataset but you only need to consider the animal classes, namely: Dog, Rooster, Pig, Cow, Frog, Cat, Hen, Insects, Sheep, Crow
- You may disregard all other classes
- You are to produce either a jupyter notebook (.ipynb) or python script (.py) to showcase your work along with a short report (.pdf) describing what you did and the results you got
- You will be evaluated based on your thought process so do include that in your report along with any other plots or visualisations

Hints:

- When dealing with audio data, there are a variety of audio features to choose from. It's best to carefully consider what these features represent and how they could be used to separate and cluster the data

- Wikipedia is a helpful resource to find out more about a particular audio feature

- Here are some resources to get you started with the variety of audio features available but note that this is by no means an exhaustive list:
    - https://towardsdatascience.com/how-i-understood-what-features-to-consider-while-training-audio-files-eedfb6e9002b
    - https://www.kaggle.com/code/ashishpatel26/feature-extraction-from-audio
    - https://devopedia.org/audio-feature-extraction

- Consider the dimension of the audio feature data and how you could best derive a representation of that data, especially if using multiple audio features

- Consider how would you visually represent the data to your stakeholders, especially if they would like to see the relationship between different classes of audio

- Although you may not use ground truth labels in your analysis, **you may use them for checking your answer**

- Librosa is a great python library for audio processing